Weekly Auto-Rebalancing Modern Efficient Portfolio System

Nicholas Christophides

Introduction

The following is a system that employs Markowitz/Modern Portfolio Theory (MPT) in order to automatically allocate or rebalance a portfolio of equities into a theoretically efficient portfolio, minimizing expected risk for a given target return. This system allows for an investor to have a portfolio following MPT without having to do any analysis or manual rebalancing. One can simply run the system once when initially investing capital, and from that point on the system will automatically perform a new analysis at the beginning of each trading week that will yield new theoretically efficient positions (without short selling) and will adjust the portfolio weights to meet these marks. Using python to automate the process and github to run the process on a fixed schedule, the result is a system allowing for autonomous, algorithmic trading with no barrier to entry. In addition to analysis and execution, a report is sent to the investor's email address detailing each rebalance as well as some risk and backtesting metrics. The purpose of this report is to detail the methodology and the reason for this system, as well as demonstrate its capabilities.

Methods

One of the largest areas of concern when using MPT for actual investment is the issue of correlations between assets. There may be correlations in the historical data used to conduct the analysis that are better ignored due to a reason such as a market crash inflating the correlation between assets higher than the true correlation. This could occur because in a market crash most equities lose value, however during normal market conditions these same assets may not move as closely together as the crash would make it seem. To account for this a technique to "de-noise" the covariance matrix between the equities analyzed is employed. This will be done using Principal Component Analysis (PCA). PCA allows for an estimate of the covariance matrix to be used in place of the sample covariance matrix over the period, ideally better representing the true covariances between equities. PCA uses a calculated amount of significant factors, which are determined to be the factors that explain most of the covariance, allowing for exclusion of the remaining factors. To determine the amount of significant factors, the Marchenko-Pastur upper bound is used as a benchmark. The Marchenko-Pastur Theorem provides a distribution for the eigenvalues of a DxN matrix, where D is the number of assets and N is the number of time periods. Upper and lower bounds

of the distribution may be calculated, representing the maximum and minimum values an eigenvalue can take if the matrix was generated randomly. Therefore, any eigenvalues above the calculated upper bound can be considered significant and used as a factor in creating the "de-noised" matrix. The upper bound of the matrix is calculated by the formula below:

$$\lambda_+ = \sigma^2 \left(1 + \sqrt{\frac{1}{q}}\right)^2 \tag{1}$$

where $q = D/N$

To estimate the covariance matrix, a factor model approach is used:

$$\Sigma = \beta\lambda\beta^T + D \tag{2}$$

where β is a p x k matrix of factor loading (the top k principal components), λ is a k x k diagonal matrix of the top k largest eigenvalues, and D is a $p \times p$ diagonal matrix representing the "noise". The resulting matrix is the estimated covariance matrix using PCA. Using the historical data gathered on the considered stocks, a covariance matrix is estimated and used in the portfolio optimization.

The process of portfolio optimization is completed by using the expected return and estimated covariance to find a portfolio which minimizes the expected volatility for a given level of expected return. In this case positions must be positive, that is, short selling is not allowed. Expected return is calculated by taking the average of the historical returns over the period considered. This quantity is updated using a rolling window approach each time the optimization is performed.

In order to determine the highest and lowest level of return to aim for, the Global Minimum Variance (GMV) portfolio must be found. In order for the list of portfolios to be efficient, there must be a portfolio with the lowest variance (variance is non-negative). Corresponding to this minimum variance portfolio, there is an expected return. Therefore, the minimum return portfolio would target the expected return of the GMV portfolio. If this were not the case, the portfolios would not be efficient, since targeting a lower return would be nonsensical as you could expect a higher return for the minimum amount of variance.

Finding the highest target return is more straightforward. To expect the highest return one would simply invest all the capital in the equity that has the highest mean return, or expected return. To solidify this concept imagine a case where half of the capital is allocated in the equity with the highest expected return, and the other half in the equity with the second highest expected return. Then the expected return would simply be the weighted average (in this case, the average) of the two expected returns. Since taking the average of a larger and smaller number yields the midpoint of the two numbers, the expected return of this portfolio would be less than if the portfolio simply consisted of the asset with the highest expected return.

Once the minimum and maximum target returns are calculated, to find an efficient frontier of portfolios one could invest in, simply find the portfolio with minimized variance for various target returns ranging from the minimum to the maximum target return. Doing so yields a curve that has portfolios along a boundary, as shown in Figure 1 below.

This curve represents something important; portfolios with expected quantities to the left of the curve are unobtainable. If this wasn't the case, then the curve wouldn't be

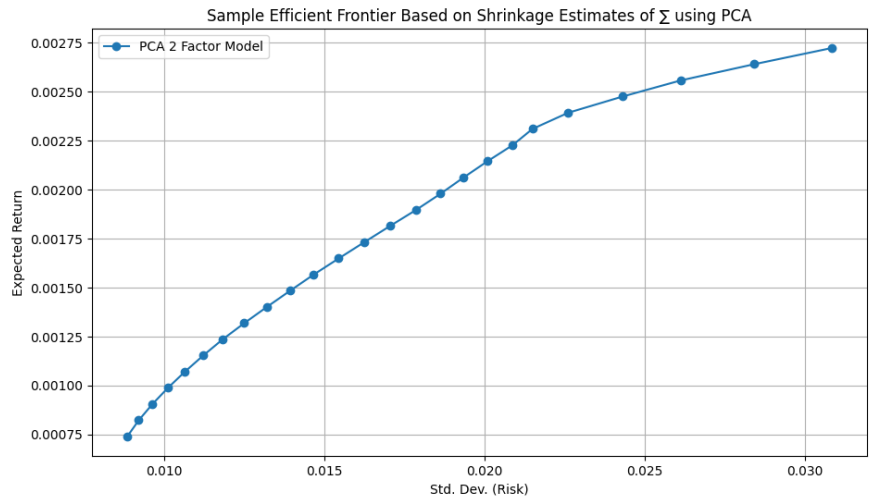

Figure 1: Sample Efficient Frontier Based on Shrinkage Estimates of Σ using PCA

efficient as one could have the same or higher level of expected return for a lower level of risk. Additionally, portfolios to the right of the curve should never be invested in, as any portfolio to the right of the curve is inefficient. That is, a portfolio landing to the right of the curve will have a higher level of risk for an expected return that can be obtained by choosing a portfolio on the curve (which all have less variance). These two facts are the reason that the curve shown above contains the theoretically efficient portfolios from the set of equities considered.

After the set of portfolios is calculated, the investor must choose a level of return they wish to target, and therefore a portfolio to adopt. This was chosen arbitrarily in this project as a return near the middle of the range of target returns, but further work should certainly include more analysis on which portfolio is right for investor preferences, or an option to choose the target return and associated risk at each rebalance. Determining the weights to achieve minimal risk for a given target return without allowing short selling is an optimization problem. The objective function is the portfolio variance. The constraints are the following;

- 1.) The sum of the weights must equal 1, that is, all allocatable capital is invested.
- 2.) The target return
- 3.) No short positions

The resulting optimization problem is shown below:

$$
\begin{gathered}
\min_{w} \quad w^\top \Sigma w \\
\text{s.t.} \\
w^\top \mu = r^* \text{ (target return)} \\
1^\top w = 1 \text{ (fully invested)} \\
0 \leq w_i \leq 1 \quad i = 1, ..., n \text{ (no shorting + no leverage)}
\end{gathered}
$$

When the portfolio is selected, then the rebalancing can occur. There are two scenarios that are possible for this system. The first case is that the capital is uninvested, that is, it is in the brokerage in cash. In this case, rebalancing isn't actually occurring. Instead the capital is simply allocated to the efficient portfolio. The second case is that the capital is invested in some combination of the assets and cash. In this case rebalancing does occur, as some positions are partially or entirely liquidated to finance the purchase of other positions, effectively "rebalancing" the portfolio to the optimal weights.

In either scenario, the system is able to execute automatically. This is done by taking the difference between the theoretically efficient position weight and the current position weight, resulting in the amount of adjustment (or rebalancing) needed for the portfolio to be theoretically efficient based on the latest data. This works because if the theoretically efficient weight is larger than the current weight the difference will be positive, meaning the system will purchase more of the asset. If the theoretically efficient weight is less than the current weight, the difference is negative, meaning the system will sell some or all of the asset to finance other positions. The sell orders are executed before the buy orders to avoid buying on margin. The proceeds from sales are used in purchasing other positions.

Some of the paramount issues when automating trades are slippage and transaction costs. To account for these, the total amount of capital available to invest for the system is less than the actual portfolio value. This is done to create a buffer against rising costs seen during slippage and transaction costs incurred on trades. A 2% to 0.5% buffer is appropriate and for larger portfolios a smaller cash buffer could be considered. Additionally, to avoid excess transaction costs and for simplicity, any position that will be adjusted by less than \$20 is not executed. This threshold is chosen arbitrarily and can be adjusted if desired.

This has been an overview of the operations and mathematics that allow this system to work. As a result of consideration of the practical issues as well as careful attention to the mathematics used in calculations, a fully automatic trading system using MPT is possible. As a summary, Figure 2 below gives a visual overview of the workflow to rebalance the portfolio.

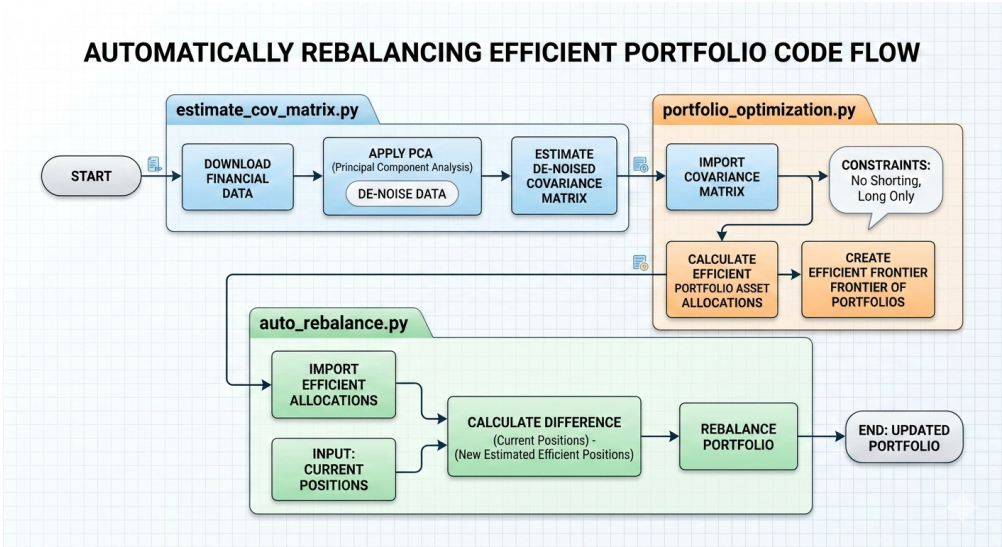

Figure 2: Workflow of the Automatic Rebalancing

Metrics and Analysis

In order to test the strategy and understand how it performs on historical data, a thorough backtest is performed and metrics of the backtest are computed. The backtest is conducted by executing the same kind of optimization that is done by the system in real time at the start of the trading week, but instead it is done given historical data only up until a point. Then, the data is moved forward by a week, the optimization is done again and the performance over the past week of data is stored. This is done for the entire backtest period, which is adjustable (150 days shown in the example). During the backtest the return sequence of the strategy is tracked and used to compute metrics. The return sequence of the strategy is plotted and compared to a simple SPY buy-and-hold strategy for comparison to a broad market index. An example of the visualization is shown below in Figure 3.

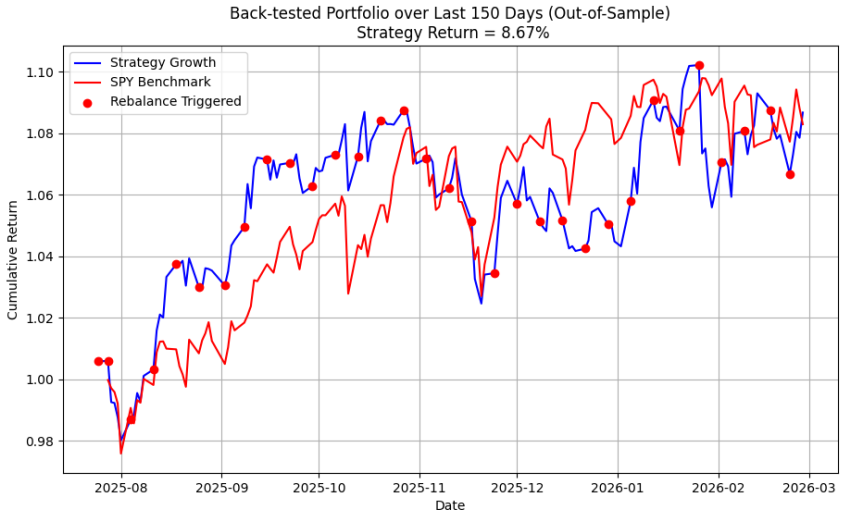

Figure 3: Backtesting of Strategy Over 150 Day Time Horizon

For a more detailed understanding of the dynamics of the strategy cumulative return, mean return, volatility, sharpe ratio, and maximum drawdown are calculated for both the strategy and the SPY benchmark over the backtest period. These methods of backtesting allow the investor to see how the portfolio would have performed over a defined period of time as well as give insight into important quantities to analyze for a portfolio to understand whether or not the strategy is performing better than a simple broad market index.

As a way of keeping the investor informed about the performance of the portfolio despite the automation of the investing, a weekly report is sent to the investor's email following each rebalancing at the start of the trading week. This report confirms that the optimization was completed without error, as well as sends the investor the metrics calculated in the 60 day backtest for both the strategy and the SPY buy-and-hold strategy for comparison. Additionally, the investor is also sent a visualization of the adjustments to the positions on the portfolio, so the investor is aware of which and how much was bought or sold of each equity during optimization to rebalance the portfolio to a theoretically efficient portfolio based on the new trading data available for the last week. An example of this visualization is pictured below in Figure 4.

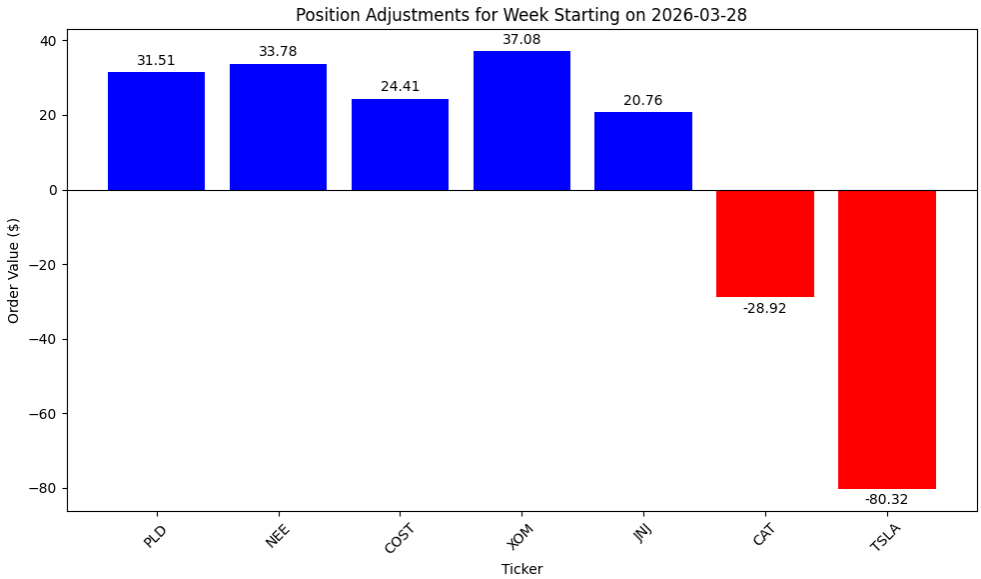

Figure 4: Example Visualization of Position Adjustments During Portfolio Rebalancing

Future Work

There is ample room for extension upon this project. The project is intended to evolve over time, with new ideas and features being integrated as time progresses. The first extension upon the system would be to conduct analysis to understand what is the optimal rebalance frequency to balance optimization and transaction costs, replacing the arbitrarily chosen weekly rebalance frequency. This could allow for more optimization while keeping transaction costs at a reasonable level. Another related concept to research and implement would be executing trades at an optimal time. If a time during the trading week and day is found to be optimal for executing trades, then the system can be modified to execute trades during this time instead of market opening at the beginning of the trading week. This could theoretically help curb slippage, or allow for more favorable prices.

The next simple extension would be to allow the investor to choose which equities or assets could be considered by the optimizer to create an efficient portfolio. This would allow for more freedom for the investor to express their preferences. Allowing the investor to choose their target return out of a certain number of calculated portfolios will also be an idea for implementation that will allow for more investor preference to be expressed. Additionally, adding more metrics to be presented to the investor such as the Value-at-Risk (VaR) of the portfolio will allow for even more insight into the behavior of the portfolio chosen by the investor and determined by the optimizer. These few ideas are some of the most obvious extensions to be integrated into the system, but there are an unlimited amount of ideas that could be used to extend the project. The project will be updated, and given enough time and change, a follow-up report may be made.

Conclusions

The system detailed in this report allows for an automated and hands-off approach for investors who are looking for a way to invest passively in the market. The automation facilitates ease of use and understanding, the report details the essential mathematics and operations needed for the optimization and automation, and the weekly email allows the investor to stay informed about their positions and how the strategy is performing compared to the SPY benchmark. Future modifications will allow for further customization of the system for the investor.

The GitHub repository containing the Python code for this project can be found at the following link: GitHub Repository